



ALMA MATER STUDIORUM
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Professor: Matteo Barigozzi

Final Project

Measuring Financial Connectedness among Emerging Equity Markets:

Diebold–Yilmaz, NETS and FNETS Compared

Marco Iacuzzi
Agustín Lancelotti
Bautista Alonso
Ezequiel Llabra

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Summary of the Base Paper and of the Network Estimators

Diebold and Yılmaz (2014) start from the observation that connectedness is central to risk measurement (market, credit and systemic risk) but is poorly defined, and that correlation-based measures capture only pairwise, linear association. They propose to measure connectedness with the variance decomposition of an approximating model: d_{ij}^H is the share of the H-step forecast-error variance of variable i due to shocks in variable j. The matrix $D^H = [d_{ij}^H]$, augmented with its row and column sums, forms a *connectedness table*. Its off-diagonal entries are pairwise directional connectedness from j to i; their row sums measure total directional connectedness *from* others to i, their column sums connectedness *to* others from j, and the difference gives net connectedness, analogous to a trade balance. Total connectedness is the average off-diagonal row (or column) sum, a single system-wide number. Because shocks are correlated, they use the generalized variance decomposition of Koop et al. (1996) and Pesaran and Shin (1998), normalized so that each row sums to one, which avoids the ordering dependence of Cholesky identification.

The paper's second contribution is conceptual: a variance-decomposition matrix is a weighted, directed network. "From" and "to" connectedness are the node from-degrees and to-degrees, and total connectedness is the mean degree of the network. The authors also relate the two directions to systemic-risk measures: from-degrees resemble exposure measures such as marginal expected shortfall, while to-degrees resemble contribution measures such as ΔCoVaR.

Empirically, they study 13 large US financial institutions from May 1999 to April 2010, using daily realized volatilities computed from five-minute intraday returns (in logs), a VAR(3) approximating model, a 12-day horizon and rolling windows of 100 days. Full-sample total connectedness is 78.3%; connectedness "from" others is tightly distributed (70–82%), whereas connectedness "to" others ranges from 53% to 106%, with Citigroup the largest net transmitter. The rolling index shows two large cycles, one from 2000 to 2003 (the end of the dot-com boom, September 11 and the WorldCom bankruptcy) and one during the 2007–2009 crisis, when it rose from about 56% in February 2007 to close to 90% in August 2007 and reached its maximum of 89.2% after the Lehman bankruptcy. Results are robust to windows of 75 and 125 days, horizons of 6 and 18 days and Cholesky orderings. Our application changes the reference universe (emerging-market country funds instead of US financial firms), the object (returns instead of volatilities) and the frequency (weekly instead of daily).

NETS (Barigozzi and Brownlees, 2019) estimates the network directly as a sparse object: a lasso-type estimator of a VAR delivers a *Granger* network (lagged dependence), the sparse concentration matrix of the VAR innovations a *contemporaneous* partial-correlation network, and their combination a *long-run* partial-correlation network. FNETS (Barigozzi et al., 2024) first removes a common factor structure, estimated by dynamic principal components in a generalized dynamic factor model (Forni et al., 2000), and then estimates the same sparse networks for the idiosyncratic component, separating connectedness due to pervasive common shocks from idiosyncratic interdependence.

1 Motivation

Emerging equity markets are exposed to two kinds of shocks. Some are global: changes in US monetary policy, in the dollar or in investors' appetite for risk move all of them at once, a pattern that Rey (2013) describes as a global financial cycle. Others are local, such as an election, a currency crisis or a change in economic policy in one country, and the question is whether, and through which countries, they spread to the rest. Telling the two apart matters for several reasons. For investors, the benefit of diversifying across emerging markets depends on how much of their co-movement is a common component that no portfolio of these markets can diversify away. For policy makers, a high degree of connectedness calls for different responses if

it reflects bilateral transmission (contagion) or a common exposure to the same external shock. For the empirical literature, the distinction is an old one: Forbes and Rigobon (2002) showed that much apparent contagion during crises is high interdependence driven by common shocks and higher volatility.

The tools to measure connectedness have multiplied, and they do not all measure the same thing. The variance-decomposition approach of Diebold and Yılmaz (2014) summarizes how much of each market's forecast-error variance is due to shocks elsewhere; it is dense by construction and does not separate common from bilateral transmission. Sparse network estimators such as NETS (Barigozzi and Brownlees, 2019) keep only the links supported by the data, and FNETS (Barigozzi et al., 2024) first removes the common factor and then estimates the network of what is left. Applied to the same data, these methods can give quite different pictures of how connected markets are, who transmits and who receives, and how stable the network is. Before using a connectedness measure to draw conclusions about contagion, it is therefore worth knowing how much of the measured connectedness depends on the estimation method.

Our sample of 17 emerging markets offers clear examples of both kinds of shock. In the week of the Argentine primary elections of August 2019 the Argentine fund (ARGT) lost 28.4% (in log terms), its largest weekly fall in the sample: a large, country-specific shock. Seven months later, the COVID-19 crash of March 2020 produced the largest weekly losses of almost every market at the same time: 14 of the 17 markets recorded their worst week in March 2020. Argentina is one case among the 17 markets in this paper, but its 2019 episode illustrates the question we ask: whether a large local shock spreads to other markets, and whether the methods agree on that.

2 Introduction

This paper asks how measured connectedness among emerging equity markets changes with the method used to estimate it, and what share of it reflects a global common factor rather than idiosyncratic links between markets. We study weekly returns of US-listed country funds for 17 emerging markets in Latin America, Asia and EMEA, together with the S&P 500 as a global reference, from March 2011 to October 2026, and we estimate connectedness in four ways: the variance-decomposition measures of Diebold and Yılmaz (2014) on a VAR estimated by OLS; the same measures on a sparse VAR estimated by the lasso with time-series cross-validation; NETS (Barigozzi and Brownlees, 2019), which estimates sparse Granger, contemporaneous and long-run partial-correlation networks; and FNETS (Barigozzi et al., 2024), which estimates the same networks after removing the common factor. All methods are estimated on the same rolling windows and mapped into comparable measures, and we compare them on three criteria: the density of the networks, the agreement between methods, and their stability over time.

We test three hypotheses. First, regularization alone should matter little for total connectedness at the weekly frequency, because the cross-market dependence of returns is mostly contemporaneous rather than lagged. Second, a substantial share of the connectedness measured by Diebold–Yilmaz should be due to a common factor, so that it falls markedly once the factor is removed, and its time variation should follow the weight of that factor. Third, sparse estimators should produce networks that are less dense, and also less stable from one window to the next, than the dense variance-decomposition network.

The results support the first two hypotheses and only partly the third. The lasso VAR sets almost all lag coefficients to zero yet gives nearly the same connectedness as Diebold–Yilmaz (80.0% against 81.0% on average over windows, correlation 0.98). NETS gives a lower level (61.5%) and keeps about half of the possible long-run links, while tracking the Diebold–Yilmaz series closely (correlation 0.89). FNETS finds a single common factor in every window, which explains on average 58% of the variance of the returns; removing it lowers connectedness by 41% on average, a share that is stable across window lengths, horizons, the exclusion of the S&P 500